

Reliable Offline Tracking for Long-Shot Windsurfing Video

Domain-specific pose, appearance, and global association

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Abstract

This report describes the engineering decisions behind a system that turns long-shot, shore-recorded windsurfing footage into identity-consistent rider tracks and stable focused views. The central challenge is not object count but error tolerance: a single identity switch can contaminate the extracted video of a rider. Large camera pans, small targets, prolonged co-visibility, and occlusions make generic online tracking a poor fit. Tracking therefore combines conservative local stitching, sail-color appearance descriptors, camera-compensated motion, and offline global tracklet linking. Once identity is established, two rig keypoints supply a stable crop anchor and rig-relative scale cue for the focused view.

1 Problem Definition and Operating Regime

1.1 The useful output is an identity-consistent video

The input is long-shot windsurfing footage recorded from shore. A typical scene contains only a handful of surfers, so detection density is modest compared with crowded pedestrian tracking. The difficulty lies elsewhere: the subjects are small, several sails can remain close or overlap for extended periods, and a rider may be partly or fully occluded before reappearing. Meanwhile, the operator follows the action with large pans—about 90° in demanding examples, and sometimes more. Almost every background feature then moves in image coordinates, while the visible background itself changes over the course of the shot.

The desired result is not merely a plausible bounding box in most frames. It is a separate, continuous trajectory for each physical surfer, suitable for extracting a focused rider video. If a track changes identity once, the resulting video jumps to another rider and can no longer serve its purpose. Identity consistency is therefore the primary objective; smooth localization and high detection recall matter only after that constraint is met.

We consequently treat association errors asymmetrically. A missed link produces two fragments that can be inspected or joined later. A false link contaminates an otherwise valid trajectory and may invalidate the complete extracted sequence. In ambiguous cases, fragmentation or abstention is preferable to an unjustified merge. This is a design requirement—effectively zero tolerance for identity switches—not a claim that the present system achieves perfect tracking on arbitrary footage.

1.2 Why ordinary online tracking is a poor fit

General-purpose multi-object trackers commonly process frames causally: they predict each active track into the next frame, assign the current detections, and carry that decision forward. This is attractive for real-time operation, but it commits at the moment evidence is weakest. During a crossing or occlusion, two continuations can be equally plausible from past motion alone; evidence that appears several seconds later cannot undo an early identity switch reliably.

Raw image-space motion is also a misleading cue in this regime. During a pan, a stationary background point and every surfer receive a large common displacement. The residual motion of an individual surfer may be much smaller than this camera component. As-

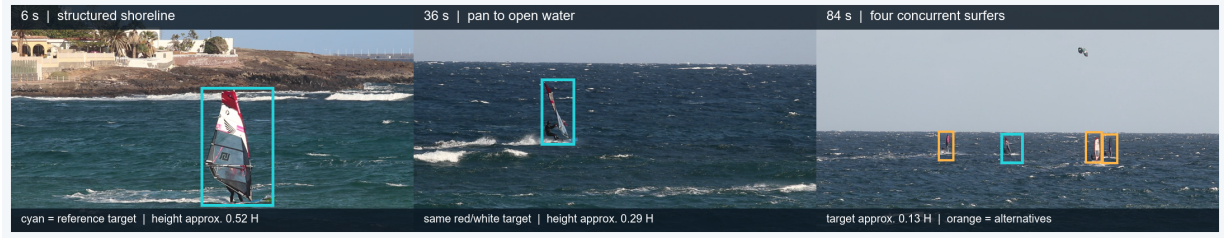


Figure 1: Representative frames from one project sequence at 6, 36, and 84 s. The cyan annotation follows the same red-and-white sail as the camera moves from shoreline to open water; its image height changes from approximately $0.52H$ to $0.13H$. At 84 s, three other surfers (orange) are visible alongside it as plausible association candidates. The composite illustrates changing background, scale variation, and identity ambiguity within one shot.

sociation therefore depends on estimating global camera motion and applying it to the motion model before comparing a prediction with a detection. The production pipeline makes this coupling explicit: it estimates masked frame-to-frame transforms while excluding detected surfers, then passes those transforms to both the greedy tracklet builder and the global linker (`video_processing/main_tracking.py`).

There is no real-time requirement for the final analysis: the complete recording is available before a focused video is generated. The system uses that freedom in two stages. A conservative causal pass accepts only clear local matches and leaves ambiguous observations as separate tracklets. An offline integer programming pass then evaluates feasible links between those tracklets jointly, using evidence on both sides of a gap. Its graph includes the option to begin or end a trajectory instead of forcing a link, and can discard very short suspected false-positive fragments in the production tracking implementation. Thus future evidence informs association without weakening the conservative local decisions that protect track purity.

The problem is therefore narrower than general multi-object tracking, but its loss function is harsher: few objects, difficult camera motion, long periods of ambiguity, and an output that is useful only while identity remains intact. The remainder of this report explains how pose, sail-specific appearance, camera compensation, and offline global association were chosen around that operating regime.

2 Domain-Specific Offline Multi-Object Tracking

The tracker is organized around the asymmetric error cost introduced in section 1. The production pipeline therefore separates association into two levels:

$$\text{detections} \longrightarrow \text{high-purity local tracklets} \longrightarrow \text{globally optimized trajectories.} \quad (1)$$

The first stage resolves only short, unambiguous transitions. The second can inspect all tracklets in the recording before deciding which gaps to bridge.

2.1 Why general-purpose components were not sufficient

The project investigated OC-SORT [1] and BoT-SORT [2], including variants with camera-motion compensation and appearance features. It also implemented appearance backends based on OSNet person re-identification and a general Vision Transformer. These are useful reference designs, but their usual operating assumptions do not match this narrow application particularly well. A causal tracker must commit while two sails overlap or cross; subsequent frames may reveal the correct continuation, but arrive too late to prevent the original identity switch. Moreover, a person-ReID network devotes its capacity to cues such as clothing and body appearance. In

long-shot windsurfing footage, the person occupies very few pixels, whereas the sail is large, deliberately colorful, and often visually distinctive.

This experience does not imply that the custom tracker is generally superior to OC-SORT, BoT-SORT, or learned ReID. The comparison is specific to shore-recorded windsurfing, offline processing, and a loss function dominated by even a single false merge. The implemented comparison and optimization tools document that the alternatives were implemented and explored, but the current benchmark harness has drifted from the production embedding and tracker interfaces. We therefore avoid a numerical outperformance claim until the baselines have been rerun on aligned inputs and representative manually verified sequences.

The design record can be summarized as follows. These are observations from project footage, not a controlled leaderboard.

Explored	Observed on this task	Chosen response
OC-SORT / BoT-SORT	Causal decisions at crossings could create unacceptable identity switches	Conservative tracklets plus offline linking
OSNet / generic ViT	Tiny riders offered weak generic identity detail; sails remained visually prominent	Sail-color descriptor
Bounding-box crop control	Rig rotation changed box centre and height, causing drift and zoom pumping	Two semantic rig keypoints
Border-based foreground model	Changing backgrounds and sails touching crop edges destabilized the estimate	Fixed centre crop margin

Table 1: The main experiment-to-design decisions recorded by the project.

2.2 Sail color as an appearance descriptor

The production appearance backend replaces generic semantic features with a compact descriptor tailored to the most stable visible property in this domain: sail color. For every detection crop it computes a joint histogram of the chromatic Lab channels (a, b) , a circular hue histogram, and a lower-weight luminance histogram. Pixel contributions are weighted by S^γ , where S is HSV saturation, so strongly colored sail panels contribute more than grey water, spray, or haze. Hue votes are linearly shared between adjacent circular bins, avoiding a discontinuity at the hue wraparound.

Color alone would discard useful coarse layout. The crop is consequently divided into three bands along its vertical axis, with a fourth block covering the complete crop. The Lab, hue, and luminance histograms are concatenated for each block. Every histogram first receives an L1 normalization and square-root mapping; the concatenated vector is then L2-normalized. The resulting representation supports a bounded Hellinger-style distance through the cosine similarity of the transformed vectors. Tracklet appearance prototypes are exponential moving averages, which reduce the effect of individual blurred, partially occluded, or poorly exposed crops.

The offline linker does not treat raw cosine overlap as a match probability. Because the descriptor consists of non-negative histogram features, two different sails can still share substantial histogram support, including residual water and background; the displayed example has a mean cross-sail cosine overlap of approximately 0.50. Production association maps the corresponding distance $d = 1 - \cos$ to the bounded heuristic score

$$p_{\text{appearance}} = \text{clip}_{[10^{-6}, 1-10^{-6}]}(1 - 11.630d). \quad (2)$$

This deliberately turns only very small descriptor distances into strong candidate evidence; it is a fitted association score, not a calibrated probability.



Figure 2: Illustrative production-descriptor comparison on three consecutive samples from one project sequence. Each matrix cell averages the pairwise appearance scores from eq. (2) over the displayed crops; diagonal cells exclude self-comparisons. Repeated observations of either sail receive strong match scores, whereas all cross-sail comparisons reach the lower clip bound. These values are neither calibrated probabilities nor benchmark accuracy. The hand-selected example explains the cue rather than evaluating it.

The descriptor also records a useful negative result. An earlier implementation estimated foreground from the color statistics of the crop border, using a robust Lab background model, a smoothness gate, and connected components. In the active path this is bypassed by a simple mask that removes the outer 10% of the crop. The fixed centre margin is less ambitious, but removes the descriptor’s dependence on a background model as the scene changes from beach to water or a sail touches the bounding-box edge. This is an example of domain adaptation through simplification: the descriptor need not segment the complete rig if its central, saturated color distribution is already sufficiently discriminative.

2.3 Conservative formation of local tracklets

Each detection initially forms a one-observation track. A greedy preprocessor then processes frames in temporal order and maintains, for every active tracklet, a constant-velocity Kalman state and an exponentially averaged appearance descriptor. Before scoring a detection, the Kalman state is advanced through the measured frame-to-frame camera transforms. Thus the expected box centre follows both the surfer’s estimated velocity and the image displacement induced by camera pan and rotation.

Candidate transitions are gated by two probabilities: one derived from the position-space Mahalanobis distance of the camera-compensated prediction, and one derived from sail-appearance distance. Strict matches require both cues to pass high thresholds and may span at most three frames. Looser candidates can remain eligible for up to six frames when supported by one strong cue or both weaker cues. Under the clean-first rule, one strict candidate wins even if loose alternatives exist; with no strict candidate, exactly one loose candidate may win. Multiple strict candidates, multiple loose-only candidates, duplicate proposals to one tracklet, or overlap with a newly created track preserve the ambiguity as separate tracklets. This deliberate strategy trades

recall for purity: uncertainty is preserved for the offline stage instead of being hidden inside an irreversible local decision.

2.4 Camera-compensated candidate costs

Let T_i and T_j be non-overlapping tracklets with T_i ending before T_j begins. The offline tracker creates a directed candidate edge $i \rightarrow j$ only when the temporal gap is non-negative and no longer than the configured horizon, currently three seconds. A Kalman filter is fitted once to all detections of T_i . Its end state is then advanced to the first observations of T_j , applying the estimated camera transform at every intervening frame. This matters during long pans: without compensation, the shared movement of the entire image would dominate the residual motion that distinguishes one surfer from another.

Every candidate receives a weighted negative-log score

$$C_{ij} = w_m C_{ij}^{\text{motion}} + w_a C_{ij}^{\text{appearance}} + w_g C_{ij}^{\text{gap}}. \quad (3)$$

The motion term averages $-\log p$ scores obtained from the Mahalanobis distance to the first two detections of T_j . The appearance term compares an exponentially weighted prototype at the end of T_i with a reverse-weighted prototype at the beginning of T_j and maps the Hellinger-style distance to a heuristic match score. The gap term is

$$C_{ij}^{\text{gap}} = -\Delta f_{ij} \log p_{\text{miss}}, \quad (4)$$

so a proposed link must accumulate stronger motion and appearance evidence as the number Δf_{ij} of unobserved frames grows. The fitted weights calibrate the relative influence of these three terms for the project data; they should not be interpreted as universal MOT constants.

The three-second horizon is a deliberate safety bound, not a claim that arbitrary occlusions can be recovered. Prolonged co-visibility and partial overlap can last much longer because detections remain available; a complete missing-observation gap beyond the horizon remains fragmented by design.

2.5 Global optimization over a bounded candidate graph

The candidate graph is solved as a minimum-cost integer program. For each tracklet i , binary variables represent starting a trajectory at i , ending at i , linking i to a candidate successor, or—for eligible short fragments—discarding i . The constraints require exactly one incoming choice and exactly one outgoing choice for every retained tracklet. The selected edges therefore form disjoint, time-ordered paths rather than competing local assignments.

Starting and ending are explicit alternatives to linking. Their costs are reduced when a tracklet begins or ends within the outer 10% of the image, because a surfer entering or leaving the field of view is more plausible there than in the centre. An edge is removed before optimization whenever linking i to j is no cheaper than ending i and starting j independently. The graph is further limited to the best few outgoing candidates per node. These operations reduce solver work while preserving the intended decision boundary: a weak link should become a fragmentation, not an identity switch.

Very short detector artifacts pose a separate problem. For fragments up to a configured number of detections, the optimizer exposes a discard option whose cost grows exponentially with fragment length. A one-frame false positive can therefore be removed without forcing it to become a trajectory or attach to a real surfer, while discarding a sustained observation rapidly becomes unattractive and is eventually disallowed.

The advantage of this formulation is non-causal rather than mystical “future knowledge.” A difficult edge is judged using a stable tracklet before the gap, a stable tracklet after it, and the other feasible links throughout the recording. The integer program selects these alternatives jointly, so two locally plausible continuations cannot both claim the same successor. The current

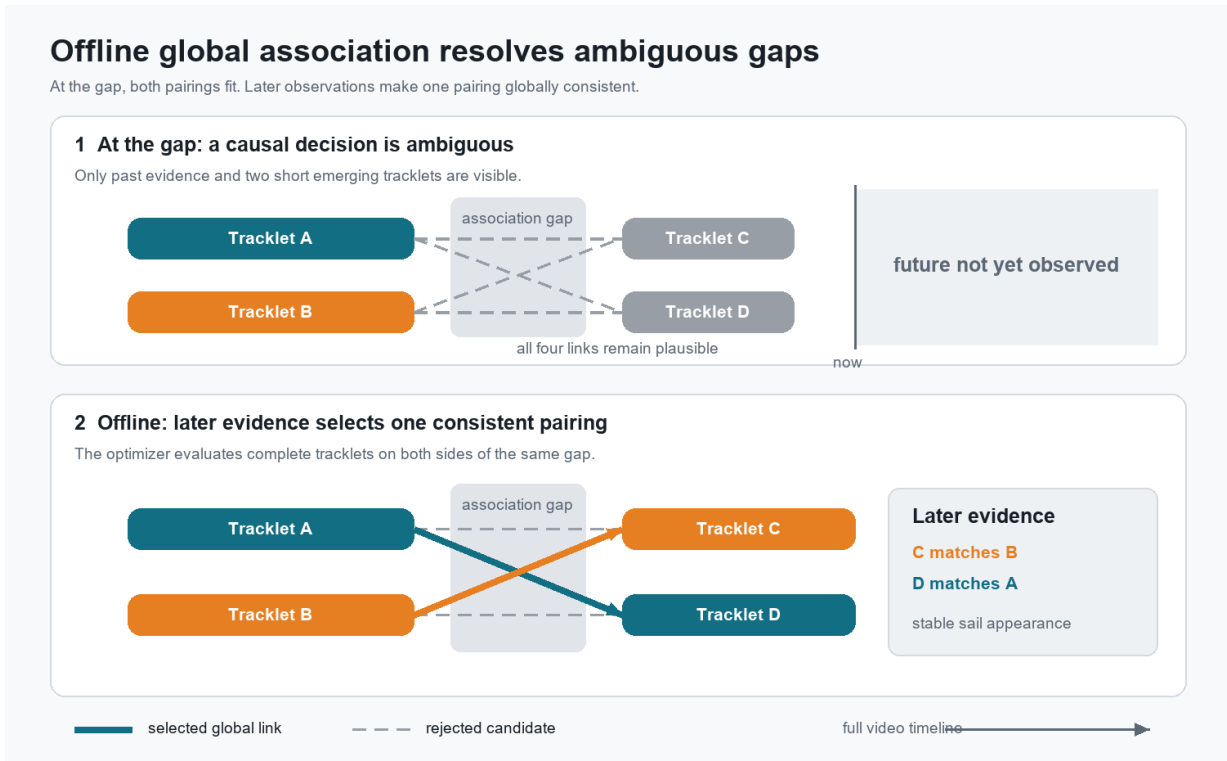


Figure 3: Why complete-video evidence helps tracklet linking. At the end of a long gap, an online tracker sees four locally plausible links between two earlier and two emerging tracklets. Offline processing can first observe the stable later portions of tracklets C and D. That later appearance evidence helps distinguish the candidate continuations, while camera-compensated motion scores the beginning of each successor. The global optimizer selects the crossed pairing A–D and B–C jointly over the bounded candidate graph. The illustration is conceptual; start, end, discard, and gap costs are omitted for clarity.

production pipeline uses this single-pass `ILPTracker`; the older iterative ILP implementation remains an experiment and is not the authoritative execution path.

2.6 From linked tracklets to renderable trajectories

The selected paths still contain sparse detections. A final post-processor removes tracks that are too short or too sparsely observed, then runs a camera-compensated forward Kalman pass and backward Rauch–Tung–Striebel smoother. It fills internal gaps with interpolated boxes and attributes before assigning compact output labels. The result is a dense, frame-indexed trajectory that the pose stage can turn into a stable crop signal without confusing interpolated observations with detector output.

3 Pose as a Stable Virtual Camera Signal

Tracking answers *which* surfer is present in each frame, but a useful focused view also needs to decide where to place the virtual camera and how tightly to crop. A detector bounding box is a poor control signal for this purpose. A windsurf rig is articulated, strongly anisotropic, and continually rotates with respect to the camera. Its axis-aligned box can therefore change width, height, and centre even when the rider has moved little and remains at approximately the same distance. Using the box centre and height directly makes the crop drift across the subject and causes visible zoom pumping.

3.1 Two semantic points instead of a generic human pose

The production detector is a YOLO pose model supervised for one class and two domain-specific keypoints: the boom–mast junction and the mast tip. These points were chosen for the rendering geometry rather than to describe the complete human body. The boom–mast junction gives a repeatable reference near the rider and sail, while the distance from that junction to the mast tip provides a rig-relative scale cue that is substantially less sensitive than an axis-aligned box height to in-plane rotation.

The labels build on the manually reviewed detection boxes. A dedicated annotation tool presents an enlarged crop for each box, permits either keypoint to be marked as not visible, and writes two-keypoint YOLO pose labels. The checked-in pose project contains 1,123 manually labelled training images and 56 validation images. The current split is an image-level development split and may contain frames from the same source sequence on both sides; a sequence-grouped held-out set is required for a generalization claim. The workflow also supports conservative pseudo-labelling: a prediction is accepted only after matching it to an existing box and applying detection, overlap, keypoint confidence, and spatial plausibility gates; pseudo-labels remain separate from manual labels and are included in training only when requested. Finally, a hard-window miner runs the complete local pipeline and ranks short intervals by keypoint jitter relative to the rendered anchor. This directs subsequent review towards failures that are visually disruptive in the final crop, rather than towards another uniform sample of easy frames.

3.2 From keypoints to crop centre

For a detection with bounding box top y_1 , height h , and centre (c_x, c_y) , the fallback anchor is

$$\mathbf{a}_{\text{box}} = (c_x, y_1 + 0.6h). \quad (5)$$

Placing it below the geometric box centre is a deliberate bias towards the rider and board rather than the empty upper part of a tall sail box. When the boom–mast junction $\mathbf{b} = (b_x, b_y)$ is visible, the implementation first constructs a low-noise proxy for the mast tip, $\tilde{\mathbf{t}} = (b_x, y_1)$, and places the semantic anchor at

$$\mathbf{a}_{\text{pose}} = 0.15\tilde{\mathbf{t}} + 0.85\mathbf{b}. \quad (6)$$

Thus the crop is anchored slightly above the boom junction. The measured mast-tip coordinate is intentionally not used for position: its larger lever arm would turn localization noise into unnecessary crop motion. It remains valuable for scale.

Small detections do not contain enough pixels for equally reliable keypoint localization. The final unsmoothed anchor therefore blends $\mathbf{a}_{\text{pose}}$ with $\mathbf{a}_{\text{box}}$ according to the shorter box dimension. The pose weight rises linearly from zero at 45 pixels to one at 55 pixels. A keypoint is considered visible at confidence 0.15 or greater; below that threshold, the box fallback is used.

Missing observations are handled as transitions, not abrupt mode switches. Gaps of up to five frames between visible boom keypoints are interpolated directly between semantic anchors. Across longer gaps, the anchor eases from the last semantic point to the per-frame box fallback and eases back when a reliable keypoint returns. Leading and trailing missing spans receive analogous fallback or short hold behaviour. A final five-frame robust weighted mean suppresses isolated jumps while giving the current frame the largest weight; Huber reweighting limits the influence of spatial outliers. This processing operates on the dense tracks produced by the tracking post-processor, so every frame in a track receives an anchor even when its detection was interpolated.

3.3 A scale tied to the rig

When both keypoints are visible, the scale signal is the Euclidean length $\ell = \|\mathbf{b} - \mathbf{t}\|_2$ of the observed boom-to-tip mast segment. Short missing spans are interpolated and the last valid value is held for up to five frames; otherwise the vertical distance from the current anchor to the



Figure 4: Pose-derived virtual-camera geometry on a representative labelled frame. The detector box and its centre (orange) describe image extent. The boom–mast junction and mast tip (cyan) instead provide a rider-relative anchor (magenta) and a rig-relative scale cue. Production rendering smooths these signals over time and falls back to box geometry when keypoints are unavailable.

box top supplies a fallback. A centred moving mean with radius 15 frames smooths this length signal. The requested crop height in pixels is then

$$C = \max\left(\frac{\bar{\ell}}{0.75}, \frac{h}{0.9}, 1\right), \quad s = \text{clip}\left(\frac{C}{H}, 10^{-6}, 1\right), \quad (7)$$

where H is the source-video height. The first term aims to make the semantic mast segment occupy 75% of the output height; the second prevents the detected box from being cropped more tightly than a 90% fill. Taking the maximum preserves context when either signal calls for a wider view.

The resulting normalized anchor and scale are stored with every renderable detection and consumed directly by the focused player. They stabilize the *composition* of an individual rider: which semantic point stays near the centre and how large the rig appears. This is distinct from global camera-motion compensation. The latter estimates frame-to-frame background transforms (while masking detected surfers), supports motion-based association, and supplies an optional stabilized overview. It does not replace the pose-derived virtual camera; a globally stabilized frame can still contain a rotating rig whose bounding box centre and dimensions fluctuate.

Table 2: Fixed-observation identity-association results on 21 development videos. “Contam.” counts predicted tracks containing more than one gold identity.

Method	Coverage	Pair P	Pair R	Pair F1	Macro F1	Contam.	Frag. excess
OC-SORT	0.981	0.985	0.662	0.792	0.857	9/165	845
BoT-SORT, no ReID	0.996	0.920	0.752	0.828	0.858	16/142	244
Production offline ILP	0.999	0.957	0.918	0.937	0.970	9/95	42

4 Evaluation

We evaluate identity association on 21 manually curated development videos. The benchmark fixes the universe of 41,079 saved, non-interpolated detections and gives every method the same bounding boxes. Of these, 41,004 belong to retained surfer identities and 75 are annotated for rejection. This deliberately isolates association from detector recall: the results answer whether the available observations were grouped correctly, not whether YOLO found every surfer.

The saved observations were selected during manual trajectory reconstruction and their boxes were Rauch–Tung–Striebel smoothed afterwards. The benchmark is therefore conditional on gold-derived observations rather than an end-to-end MOT test. It is also an in-sample development-set evaluation: algorithm design was informed by these videos, and the provenance of the original ILP parameter tuning is not fully recorded. We report the results as engineering evidence and failure analysis, not as a held-out leaderboard.

4.1 Metrics

For retained detections, pairwise precision and recall compare every unordered pair. A pair is positive when both observations have the same annotated identity:

$$P_{\text{pair}} = \frac{\text{correct same-identity pairs}}{\text{all pairs joined by the tracker}}, \quad (8)$$

$$R_{\text{pair}} = \frac{\text{correct same-identity pairs}}{\text{all annotated same-identity pairs}}. \quad (9)$$

False merges reduce precision and fragmentation reduces recall. Missing outputs are treated as singleton predictions rather than being silently omitted. Micro scores pool all pairs and consequently weight long identities quadratically; macro scores average per-video values. We additionally report contaminated predicted tracks, which contain more than one gold identity; fragmentation excess, the number of predicted pieces beyond one per gold identity; and exact videos, whose complete partition matches the annotation up to track-label permutation. These absolute counts matter because a high aggregate F1 can conceal one application-breaking identity switch.

4.2 Fixed-observation comparison

Table 2 compares the production pipeline with OC-SORT and BoT-SORT. OC-SORT receives the fixed detections directly. BoT-SORT uses its ECC camera compensation but no pedestrian-specific ReID network. The production method reruns conservative preprocessing, masked video stabilization, and offline ILP association. Its sail descriptor uses border-derived foreground masking and a whole-fragment appearance prototype; motion gating evaluates centre, width, and height; and implausible motion, area, or aspect-ratio transitions are excluded before optimization.

The production system gives the strongest overall association score and reduces fragmentation by a factor of approximately six relative to BoT-SORT and by a factor of twenty relative to OC-SORT. Eleven of 21 videos are exact, compared with four for BoT-SORT and one for OC-SORT.

Table 3: Staged production-pipeline revision on the same development set.

Configuration	Pair P	Pair R	Pair F1	Contam.	Frag.
Original reconstruction	0.907	0.880	0.893	20	48
Mask + whole-fragment appearance + 4D motion	0.939	0.917	0.928	17	44
+ conservative link vetoes	0.957	0.918	0.937	9	42

OC-SORT attains the highest pairwise precision by remaining highly conservative, but its 845 excess fragments do not provide the continuous identity track required by the application. The production result is a more useful operating point for this dataset, but its nine contaminated tracks mean that it still does not satisfy the nominal zero-false-merge requirement.

Runtime is included only as an operational diagnostic. On the evaluation machine, OC-SORT took 9.0 seconds, BoT-SORT 497.7 seconds, and the production association path 402.3 seconds. The workloads are not identical: OC-SORT needs no video decode, BoT-SORT includes decode and ECC, and production includes masked stabilization but excludes the initial crop-embedding extraction.

4.3 Safeguard ablation and failure analysis

The initial production reconstruction exposed a concentrated false-link problem. Preprocessor fragments were almost always identity-pure, but the ILP sometimes joined two pure fragments from different surfers. Many of the unexpected short-gap errors occurred at crossings, where the endpoint crops contained both sails. The previous appearance model emphasized precisely those boundary frames, the color descriptor’s intended foreground mask was bypassed by an early return, and the ILP requested a position-only projection from a Kalman model that already represented width and height.

Table 3 records two staged revisions. Because several changes are bundled in the first stage, this is a system ablation rather than a claim that any single component produced the full difference. Foreground masking, whole-fragment appearance aggregation, and four-dimensional motion evaluation raise micro F1 from 0.893 to 0.928. Adding conservative vetoes for motion NLL and threefold area or aspect-ratio discontinuities raises precision further and halves contaminated tracks, without increasing fragmentation.

The vetoes are not a second tracker layered on top of the Kalman filter. The Kalman state already contains centre, width, and height; the earlier ILP call discarded the size dimensions. The revised system uses the full four-dimensional Mahalanobis evidence and additionally refuses links whose absolute evidence is implausible. This second step is necessary because the ILP’s start and end penalties can otherwise make even a very poor link cheaper than splitting two central fragments.

A manual audit of the remaining selected links found nine wrong fragment joins. The residual cases are qualitatively different from the removed geometric failures. They mostly involve crossings, nearly identical sails, or a fragment whose crop contains both competitors; their position, scale, and often color evidence are genuinely plausible. We tested abstention based only on the objective gap to the best solution with one selected edge removed. That margin did not separate errors: at useful thresholds it rejected substantially more correct links than wrong ones. We therefore do not add a development-set-tuned global margin rule. Resolving the remaining cases would require an overlap-aware appearance model or explicit paired-crossing logic, rather than another scalar threshold.

4.4 Interpretation and limitations

The comparison supports the report’s central engineering claim: an offline, domain-specific association model gives a substantially better continuity/contamination trade-off for these long-shot

windsurfing videos than the evaluated general-purpose trackers. It does not support a claim of state-of-the-art MOT performance or perfect tracking. The benchmark omits detector errors, uses development data, and includes gold-derived smoothed boxes. BoT-SORT is also evaluated without its pedestrian ReID model because that embedding is mismatched to sails.

The machine-readable per-video results, exact tracker configurations, and reproduction command are released with the repository. Product acceptance still requires complete-trajectory review: the reported pair metrics summarize millions of pairs, whereas one remaining identity switch can invalidate the final crop for a user.

5 Frame-Accurate Delivery

Identity-consistent tracks are useful only if the browser applies metadata to the frame that produced it. Ordinary HTML video exposes media time rather than stable frame identity, so the WebCodecs path [3] defines a canonical index by ordering packets by presentation timestamp and sequence number. Preview seeks from a preceding keyframe and decodes forward. Export decodes independently but maps each sample to the same index, resynchronizing by timestamp when necessary. The shared index and crop geometry preserve frame correspondence through preview and export; this is a delivery constraint rather than a tracking contribution.

6 Lessons and Limitations

The main lesson is not that custom machinery should replace general-purpose tracking, but that a narrow output and asymmetric failure cost can justify a narrower system. Conservative tracklets preserve uncertainty; sail color and camera-compensated motion score continuations; and offline optimization chooses compatible links jointly. Two rig keypoints separately stabilize focused-view composition.

No input-aligned held-out benchmark shows that this system is numerically superior to the general-purpose trackers explored here. Similar sails, small targets, and feature-poor backgrounds can weaken the cues; gaps beyond the three-second horizon remain fragmented by design. Measuring the near-zero identity-switch objective requires aligned evaluation and complete-track audits. Within those limits, every component follows one contract: preserve identity, stabilize the rider-relative view, and maintain exact frame correspondence through preview and export.

References

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